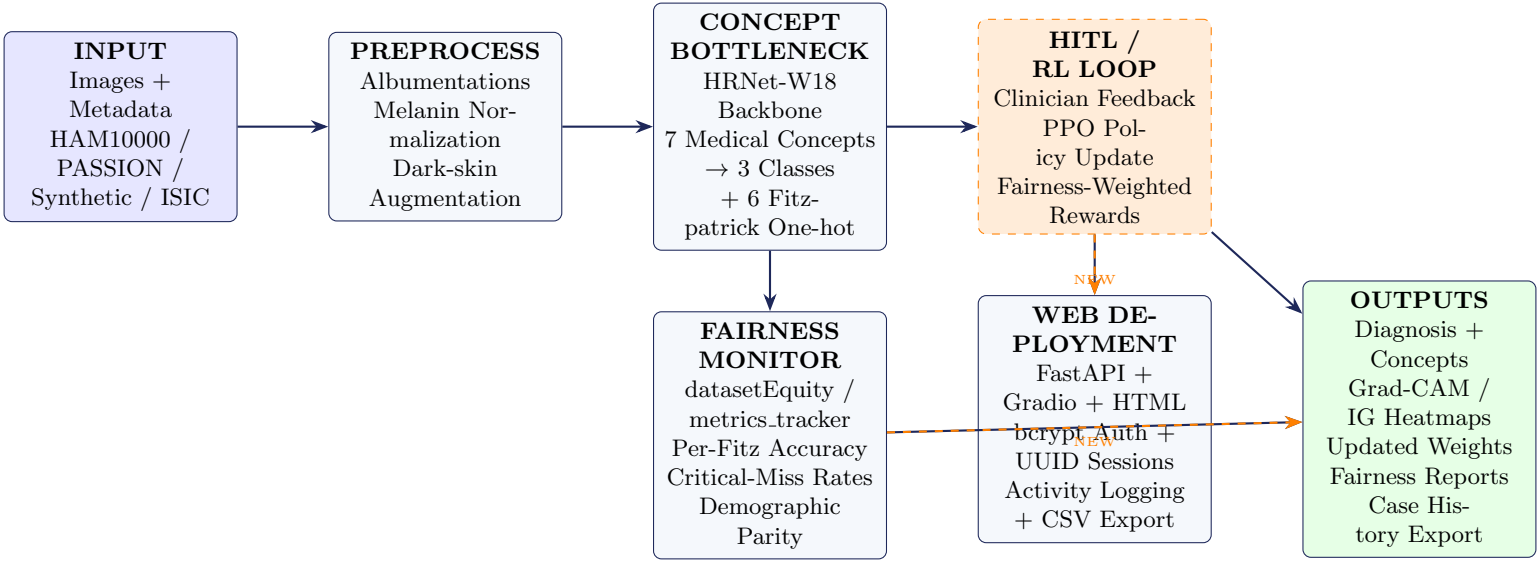
INPUT
Images + Metadata
HAM10000 / PASSION / Synthetic / ISIC
PREPROCESS
Albumentations
Melanin Normalization
Dark-skin Augmentation
CONCEPT BOTTLENECK
HRNet-W18 Backbone
7 Medical Concepts → 3 Classes
+ 6 Fitzpatrick One-hot
HITL / RL LOOP
Clinician Feedback
PPO Policy Update
Fairness-Weighted Rewards
NEW
FAIRNESS MONITOR
datasetEquity / metrics_tracker
Per-Fitz Accuracy
Critical-Miss Rates
Demographic Parity
WEB DEPLOYMENT
FastAPI + Gradio + HTML
bcrypt Auth + UUID Sessions
Activity Logging + CSV Export
NEW
OUTPUTS
Diagnosis + Concepts
Grad-CAM / IG Heatmaps
Updated Weights
Fairness Reports
Case History Export